The documentation and process conversion measures necessary to comply with this document shall be completed by 10 April, 2003.

INCH-POUND

MIL-PRF-19500/485H
10 January 2003
SUPERSEDING
MIL-PRF-19500/485G
7 August 2001

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, LOW-POWER
TYPES: 2N5415, 2N5415S, 2N5415UA, 2N5416, 2N5416S, AND 2N5416UA
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for PNP, silicon, switching transistors. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500. Two levels of product assurance are provided for die.

1.2 Physical dimensions. See figure 1 (similar to T0- 5), and 2, 3, and 4 for JANHC and JANKC (die) dimensions, and figure 5 for UA package

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Types	P_T (1) T_A = +25°C	P_T (2) T_C = +25°C	V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_{STG} and T_{OP}	$R_{\theta JA}$	$R_{\theta JC}$
	W	W	V dc	V dc min	V dc	A dc	°C	°C/W	°C/W
2N5415, S	0.75	10	200	200	6.0	1.0	-65 to +200	234	17.5
2N5415UA	0.75	10	200	200	6.0	1.0	-65 to +200	234	17.5
2N5416, S	0.75	10	350	300	6.0	1.0	-65 to +200	234	17.5
2N5416UA	0.75	10	350	300	6.0	1.0	-65 to +200	234	17.5

(1) Derate linearly 4.28 mW/°C for T_A > +25°C.
(2) Derate linearly 57.1 mW/°C for T_C > +25°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC-VAC, Post Office Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Primary electrical characteristics.

	h_{FE1} I_C = 50 mA dc, (1) V_{CE} = 10 V dc	C_{obo} I_E = 0, V_{CB} = 10 V dc, 100 kHz ≤ f ≤ 1 MHz	$\lvert h_{fe} \rvert$ I_C = 10 mA dc V_{CE} = 10 V dc, f = 5 MHz	V_{BE} I_C = 50 mA dc, (1) V_{CE} = 10 V dc	$V_{CE(sat)}$ I_C = 50 mA dc, (1) I_B = 5 mA dc
		pF		V dc	V dc
Min	30		3		
Max	120	15	15	1.5	2.0

1/ Pulsed (See 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

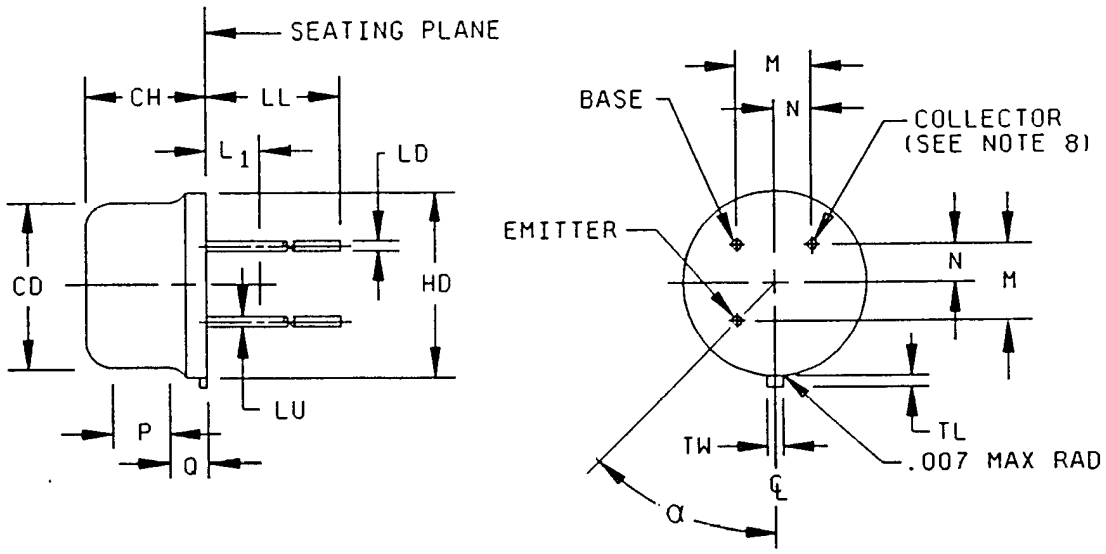
DEPARTMENT OF DEFENSE

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Services (DAPS), Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
HD	.335	.370	8.51	9.40	
CD	.305	.335	7.75	8.51	
CH	.240	.260	6.10	6.60	
LL	See notes				10,12,13
L1		.050		1.27	11
LD	.016	.021	0.41	0.53	3,10
LU	.016	.019	0.41	0.48	4,10

Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
P	.100		2.54		5
Q					6
TL	.029	.045	0.74	1.14	9
TW	.028	.034	0.71	0.86	
a	45° TP				7
M	.1414 Nom		3.59 Nom		7
N	.0707 Nom		1.80 Nom		7

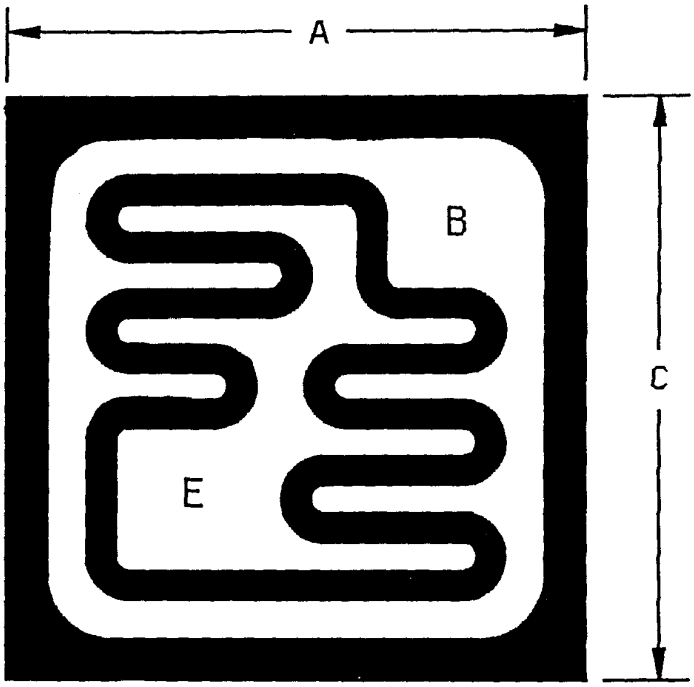
FIGURE 1. Physical dimensions (similar to T0- 5).

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Measured in the zone beyond .250 inch (6.35 mm) from the seating plane.
4. Measured in the zone .050 inch (1.27 mm) and .250 inch (6.35 mm) from the seating plane.
5. Variations on dimension CD in this zone shall not exceed .010 inch (0.25 mm).
6. Outline in this zone is not controlled.
7. When measured in gauging plane .054 +.001, -.000 inch (1.37 +0.03, -0.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 inch (0.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 2 shows the preferred method of measurement.
8. The collector shall be electrically connected to the case.
9. Measured from the maximum diameter of the actual device.
10. All 3 leads (see 3.5.1).
11. Diameter of leads in this zone is not controlled.
12. For transistor types 2N5415 and 2N5416, dimension LL shall be 1.500 inches (38.10 mm) minimum and 1.75 inches (44.5 mm) maximum.
13. For transistor types 2N5415S and 2N5416S, dimension LL shall be .5 inch (12.70 mm) minimum and 0.75 inch (19.05 mm) maximum.
14. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions - Continued.

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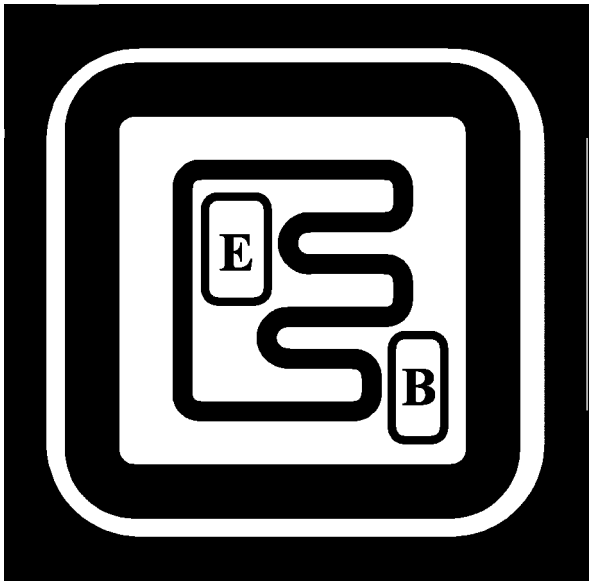
A version

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.039	.043	0.99	1.09
C	.039	.043	0.99	1.09

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
3. The physical characteristics of the die are:
   Thickness is .006 inch (0.15 mm) minimum, .012 inch (0.30 mm) maximum.
   Top metal: Aluminum 17,500 Å minimum, 20,000 Å nominal
   Back metal: Gold 2,500 Å minimum, 3,000 Å nominal.
   Back side: Collector; Bonding pad: B = .004 inch (0.10 mm) x .005 inch (0.13 mm).
   E = .004 inch (0.10 mm) x .0055 inch (0.14 mm).

FIGURE 2. JANHC and JANKC (A-version) die dimensions.

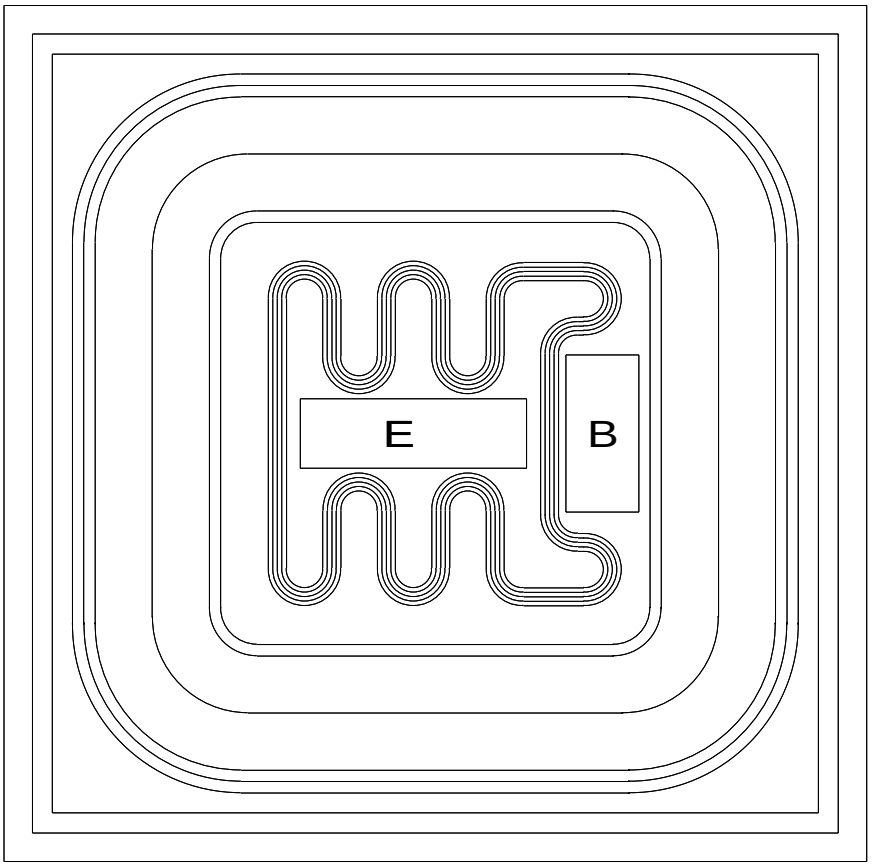
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1. Chip size: .046 x .046 inch ±.002 inch (1.1684 x 1.1684 mm ±.0508 mm).
2. Chip thickness: .010 ±.0015 inch (.254 ±.0381 mm) nominal.
3. Top metal: Aluminum 15,000Å minimum, 18,000Å nominal.
4. Back metal: A. Al/Ti/Ni/Ag 12kÅ/3kÅ/7kÅ/7kÅ minimum, 15kÅ/5kÅ/10kÅ/10kÅ nominal.
   B. Gold 2,500Å minimum, 3000Å nominal.
5. Backside: Collector.
6. Bonding pad: B = .005 x .008 inch (.127 x .2032 mm), E = .010 x .007 inch (.254 x .1778 mm).

FIGURE 3. JANHCB and JANKCB (B-version) die dimensions.

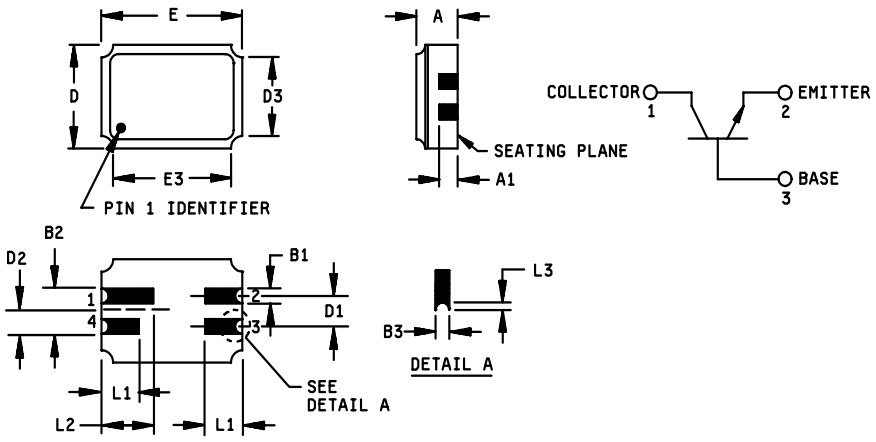
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Die size:	.045 x .045 inch (1.143 x 1.143 mm).
Die thickness:	.008 ±.0016 inch (0.2032 ±0.04064 mm).
Base pad:	.0085 x .00425 inch (0.2159 x 0.10795 mm).
Emitter pad:	.004 x .0125 inch (0.1016 x 0.3175 mm).
Back metal:	Gold, 6,500 ±1,950 Ang.
Top metal:	Aluminum, 20,000 ±2,000 Ang.
Back side:	Collector.
Glassivation:	SiO_2, 7,500 ±1,500 Ang.

FIGURE 4. JANHCD and JANKCD (D-version) die dimensions.

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Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.061	.075	1.55	1.90	3
A1	.029	.041	0.74	1.04	
B1	.022	.028	0.56	0.71	
B2	.075 REF		1.91 REF		
B3	.006	.022	0.15	0.56	5
D	.145	.155	3.68	3.93	
D1	.045	.055	1.14	1.39	
D2	.0375 BSC		.952 BSC		
D3		.155		3.93	
E	.215	.225	5.46	5.71	
E3		.225		5.71	
L1	.032	.048	0.81	1.22	
L2	.072	.088	1.83	2.23	
L3	.003	.007	0.08	0.18	5

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension "A" controls the overall package thickness. When a window lid is used, dimension "A" must increase by a minimum of .010 inch (0.254 mm) and a maximum of .040 inch (1.020 mm).
4. The corner shape (square, notch, radius, etc.) may vary at the manufacturer's option, from that shown on the drawing.
5. Dimensions "B3" minimum and "L3" minimum and the appropriately castellation length define an unobstructed three-dimensional space traversing all of the ceramic layers in which a castellation was designed. (Castellations are required on bottom two layers, optional on top ceramic layer.) Dimension "B3" maximum and "L3" maximum define the maximum width and depth of the castellation at any point on its surface. Measurement of these dimensions may be made prior to solder dipping.

FIGURE 5. Physical dimensions, surface mount (UA version).

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3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-19500.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1, 2, 3, 4, and 5 herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4 VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 JANHC and JANKC die. Qualification for die shall be in accordance with MIL-PRF-19500.

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* 4.3 Screening (JANS, JANTXV, and JANTX levels only). Screening shall be in accordance with table IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c	Thermal impedance method 3131 of MIL-STD-750.	Thermal impedance method 3131 of MIL-STD-750.
9	I_{CEX} and h_{FE1}	Not applicable
10	48 hours minimum	48 hours minimum
11	I_{CEX} , h_{FE1} ΔI_{CEX} = 100 percent of initial value or 2.5 µA dc, whichever is greater Δh_{FE1} = ± 15 percent of initial value.	I_{CEX} and h_{FE1}
12	See 4.3.1 240 hours minimum	See 4.3.1 80 hours minimum
13	Subgroups 2 and 3 of table I herein; ΔI_{CEX} = 100 percent of initial value or 2.5 µA dc, whichever is greater; Δh_{FE1} = ±15 percent of initial value.	Subgroup 2 of table I herein; ΔI_{CEX} = 100 percent of initial value or 2.5 µA dc, whichever is greater; Δh_{FE1} = ± 20 percent of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: V_{CB} = 10 - 30 Vdc. Power shall be applied to achieve T_J =135°C minimum using a minimum P_D = 75 percent of P_T maximum rated as defined in 1.3.

* 4.3.2 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500, “Discrete Semiconductor Die/Chip Lot Acceptance”. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

4.3.3 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with method 3131 of MIL-STD-750.

a. I_H forward heating current ----------200 mA to 330 mA.

b. t_H heating time ----------30 ms.

c. I_M measurement current----------5 mA min.

d. t_{md} measurement delay time -------60 µs max.

e. V_{CE} collector-emitter voltage -------10 V dc.

f. V_{CB} collector-base voltage ----------10 V dc.

The maximum limit for $Z_{\theta JX}$ under these test conditions are $Z_{\theta JX}$ (max) = 35°C/W.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed per MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied per 4.4.2)

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table VIa (JANS) of MIL-PRF-19500 and 4.4.2.1 herein. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, subgroup 2 and 4.5.3 herein. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with table I, subgroup 2 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

	Subgroup	Method	Conditions
*	B4	1037	V_{CB} = 10 V dc; 2,000 cycles.
*	B5	1027	(Note: If a failure occurs, resubmission shall be at the test conditions of the original sample.) V_{CB} = 10 V dc, $P_D \geq$ 100 percent of maximum rated P_T (see 1.3). Option 1: 96 hours minimum sample size in accordance with table VIa of MIL-PRF-19500, adjust T_A or P_D to achieve T_J = +275°C minimum. Option 2: 216 hours minimum, sample size = 45, c = 0; adjust T_A or P_D to achieve T_J = +225°C minimum.
	B6	3131	See 4.5.2.

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new “assembly lot” option is exercised, the failed assembly lot shall be scrapped.

Step	Method	Condition
1	1039	Steady-state life: Test condition B, 340 hours, V_{CB} = 10 - 30 V dc, power shall be applied to achieve T_J = +150°C minimum using a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0.
2	1039	The steady-state life test of step 1 shall be extended to 1,000 hours for each die design. Samples shall be selected from a wafer lot every twelve months of wafer production. Group B, step 2 shall not be required more than once for any single wafer lot. n = 45, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0.

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4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, subgroup 2 herein.

	Subgroup	Method	Conditions
	C2	2036	Test condition E, except the UA package.
	C5	3131	See 4.5.2.
*	C6	1026	V_{CB} = 10 - 30 V dc. Power shall be applied to achieve T_J = +150°C minimum using a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3.

4.4.3.1 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

* 4.4.4 Group E inspection. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the associated specification that did not request the performance of table II tests, the tests specified in table II herein must be performed to maintain qualification.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following conditions shall apply: Maximum $R_{\theta JC}$ shall be 17.5°C/W.

a. I_H: Collector heating current..... 0.2 A.

b. t_H: Heating time..... Steady-state.

c. I_M: Measurement current..... 10 mA dc.

d. t_{MD}: Measurement delay time ..... 60 µs.

e. t_{SW}: Sampling window time ..... 10 µs maximum.

f. V_{CE}: Measurement current (same as V_H)..... 25 V dc.

g. V_H: Collector-emitter heating voltage ..... 25 V dc.

4.5.3. Delta requirements. Delta requirements shall be as specified below:

Step	Inspection	MIL-STD-750		Symbol	Limit	Unit
		Method	Conditions			
1	Collector-emitter cutoff current	3041	Bias condition D, See table I for conditions	ΔI_{CE01} See note (1)	100 percent of initial value or 500 nA dc, whichever is greater.	
2	Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 50 mA dc; pulsed see 4.5.1	Δh_{FE1} See note (1)	±25 percent change from initial reading.	

(1) Devices which exceed the group A limits for this test shall not be accepted.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical 3/ examination	2071	n = 45 devices, c = 0				
Solderability 3/ 4/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/ 4/ 5/	1022	n = 15 devices, c = 0				
Temp cycling 3/ 4/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Hermetic Seal 4/ Fine leak Gross leak	1071	n = 22 devices, c = 0				
Electrical measurements 4/		Table I, subgroup 2				
Bond strength 3/ 4/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = 300°C at t = 2 hrs n = 11 wires, c = 0				
Decap internal visual design verification	2075	n = 4, c = 0.				
Subgroup 2						
Collector to emitter cutoff current	3041	Bias condition D	I_{CE01}		50	µA dc
2N5415,S 2N5416,S		V_{CE} = 150 V dc V_{CE} = 250 V dc				
Collector to emitter cutoff current	3041	Bias condition D	I_{CE02}		1	mA dc
2N5415,S 2N5416,S		V_{CE} = 200 V dc V_{CE} = 300 V dc				
Collector to emitter cutoff current	3041	Bias condition D, V_{BE} = 1.5 V dc	I_{CEX}		50	µA dc
2N5415,S 2N5416,S		V_{CE} = 200 V dc V_{CE} = 300 V dc				
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 6 V dc	I_{EBO}		20	µA dc
Collector to base cutoff current	3036	Bias condition D	I_{CBO1}		50	µA dc
2N5415,S 2N5416,S		V_{CB} = 175 V dc V_{CB} = 280 V dc				

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup2 - continued						
Collector to base cutoff current 2N5415,S 2N5416,S	3036	Bias condition D V_{CB} = 200 V dc V_{CB} = 350 V dc	I_{CBO2}		500	µA dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 50 mA dc, pulsed (see 4.5.1)	h_{FE1}	30	120	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 1 mA dc, pulsed (see 4.5.1)	h_{FE2}	15		
Collector to emitter voltage (saturated)	3071	I_C = 50 mA dc, I_B = 5 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)}$		2.0	V dc
Base emitter voltage saturation	3066	Test condition B, V_{CE} = 10 V dc, I_C = 50 mA dc, pulsed (see 4.5.1)	V_{BE}		1.5	V dc
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to base cutoff current 2N5415,S 2N5416,S	3036	Bias condition D V_{CB} = 175 V dc V_{CE} = 280 V dc	I_{CB03}		1	mA dc
Low temperature operation:		T_A = -65°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 50 mA dc, pulsed (see 4.5.1)	h_{FE3}	15		
Subgroup 4						
Small-signal short- circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc, I_C = 10 mA dc, f = 5 MHz	$\lvert h_{fe} \rvert$	3	15	
Small-signal short- circuit forward-current transfer ratio	3206	V_{CE} = 10 V dc, I_C = 5 mA dc, f ≤ 1 kHz	h_{fe}	25		
Open circuit output capacitance	3236	V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{obo}		15	pF
Input capacitance(output open-circuited)	3240	V_{EB} = 5 V dc, I_C = 0, 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		75	pF
Subgroup 5						
Pulse response	3251	Test condition A				
Turn-on time		V_{CC} = 200 V dc, I_C = 50 mA dc, I_{B1} = 5 mA dc, see figure 6.	t_{on}		1	µs
Turn-off time		V_{CC} = 200 V dc, I_C = 50 mA dc, I_{B1} = I_{B2} = 5 mA dc, see figure 6.	t_{off}		10	µs

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 6						
Not applicable						
Subgroup 7						
Breakdown voltage, collector to emitter	3011	I_C = 50 mA dc, I_B = 5 mA dc, L = 25 mH; f = 30 - 60 Hz (see figure 7)	$V_{(BR)CEO}$			
2N5415, S				200		V dc
2N5416, S				300		V dc
Safe operating area (continuous dc)	3051	T_C = +25°C, 1 cycle, t = 0.4 s, (see figure 8). n = 22, c = 0				
Test 1		V_{CE} = 10 V dc, I_C = 1 A dc				
Test 2		V_{CE} = 100 V dc, I_C = 100 mA dc				
Test 3 (2N5415, S only)		V_{CE} = 200 V dc, I_C = 24 mA dc				
Test 4 (2N5416, S only)		V_{CE} = 300 V dc, I_C = 10 mA dc				
Electrical measurements		See table I, subgroup 2 herein.				

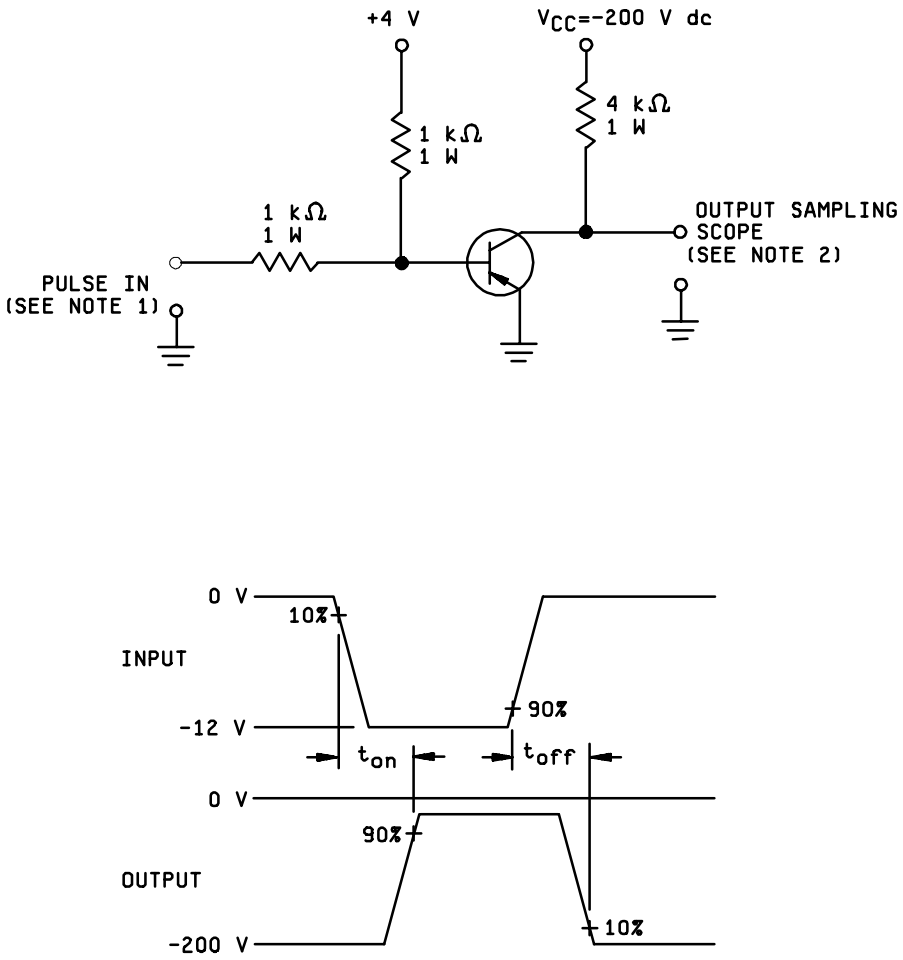
1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed subgroup A1, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.

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TABLE II. Group E inspection (all quality levels) - For qualification only

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
* Subgroup 1			
Temperature cycling (air to air)	1051	Test condition C, 500 cycles	45 devices c = 0
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2 herein.	
* Subgroup 2			
Intermittent life	1037	V_{CB} = 10 V dc, 6,000 cycles.	45 devices c = 0
Electrical measurements		See table I, subgroup 2 herein.	
Subgroups 3, 4, 5, 6 and 7			
Not applicable			
* Subgroup 8			
Reverse stability	1033	Condition A for devices ≥ 400 V dc. Condition B for devices < 400 V dc.	45 devices c = 0

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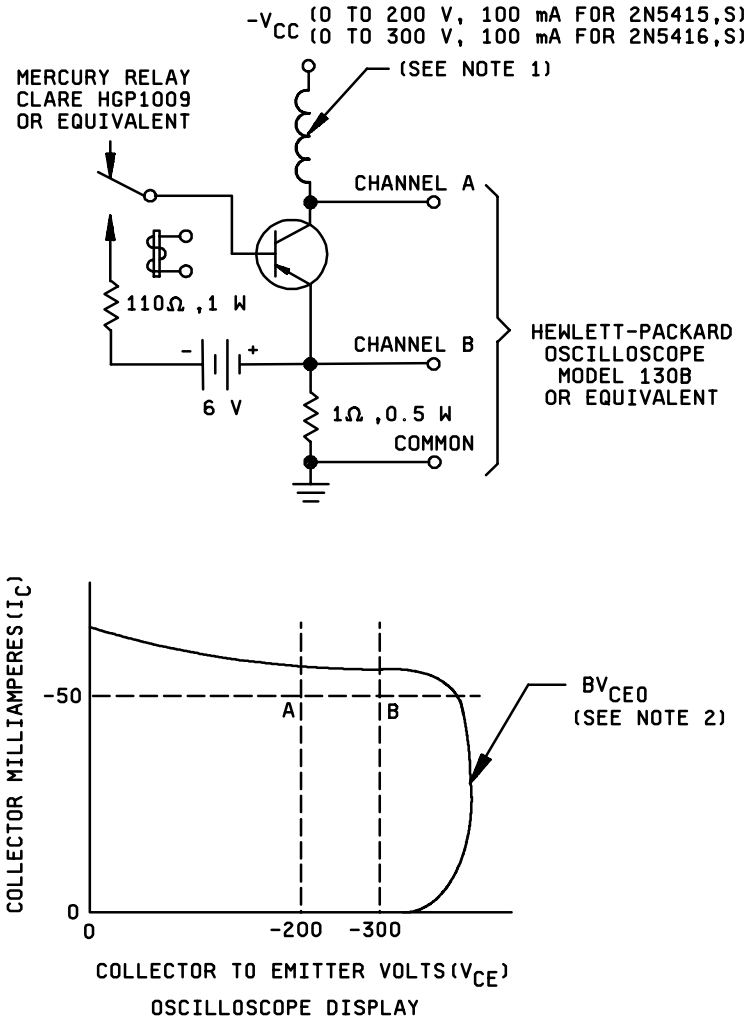


NOTES:

1. The rise time (t_r) and fall time (t_f) of applied pulse shall be each < 20 ns; duty cycle < 2 percent; generator source impedance shall be 50 Ω; pulse width = 20 μs.
2. Output sampling oscilloscope: Z_{in} > 100 kΩ; C_{in} < 50 pF; rise time < 2 ns.

FIGURE 6. Pulse response test circuit.

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NOTES:

1. L = 25 mH, Q = 76, R = 83.4 Ω, I = 75 mA (J.W. Miller number 6308 or equivalent).
2. BV_{CEO} is acceptable when the trace falls to the right and above point “A” for type 2N5415. The trace shall fall to the right and above point “B” for type 2N5416.

FIGURE 7. BV_{CEO} measurement test circuit.

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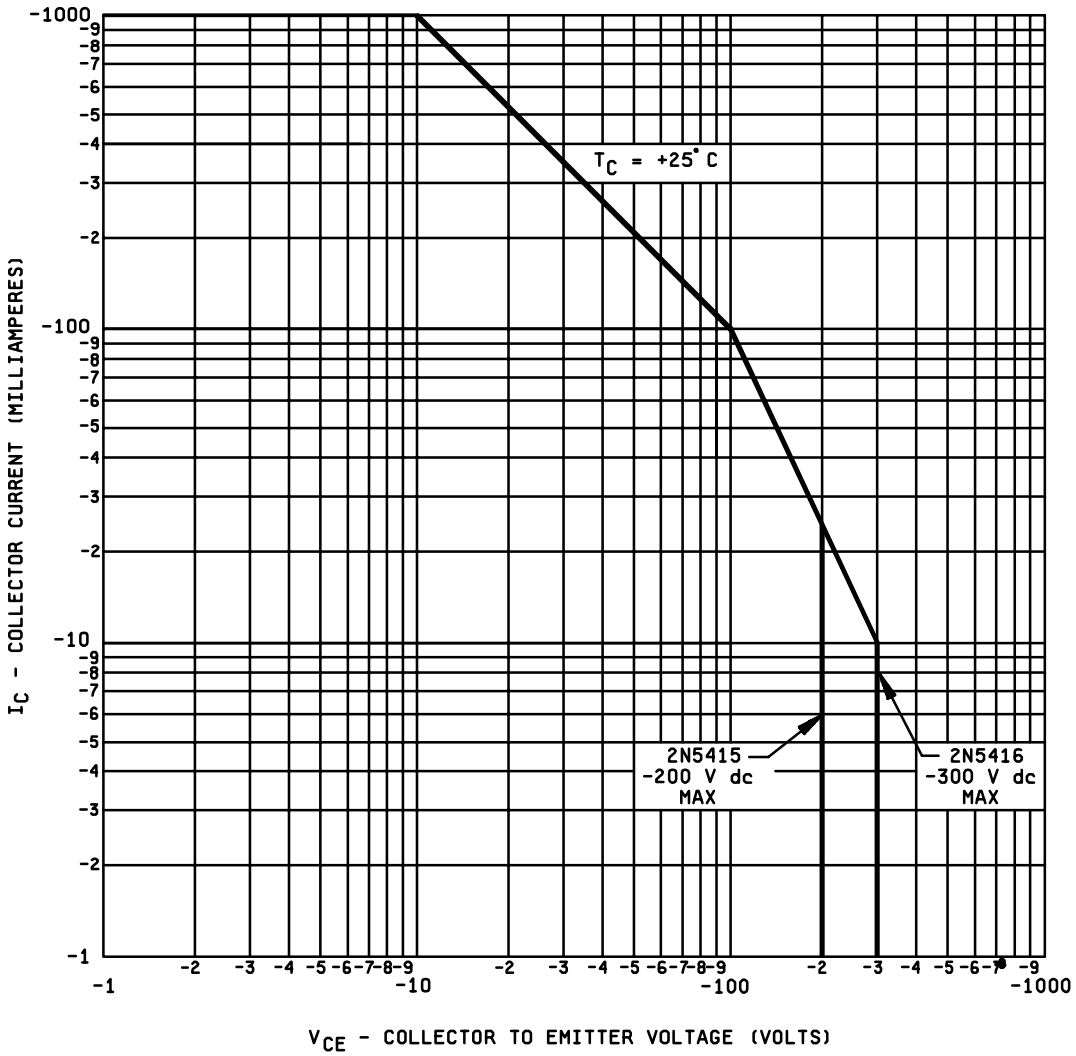


FIGURE 8. Maximum safe operating graph (continuous dc).

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5 PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2.1).

c. Packaging requirements (see 5.1).

d. Type designation and product assurance level.

e. Lead finish (see 3.4.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43216-5000.

6.4 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example, JANHCA2N5415) will be identified on the QPL.

JANC ordering information			
PIN	Manufacturers CAGE		
	34156	33178	43611
2N5415	JANHCD2N5415 JANKCD2N5415	JANHCA2N5415 JANKCA2N5415	JANHCB2N5415 JANKCB2N5415
2N5416	JANHCD2N5416 JANKCD2N5416	JANHCA2N5416 JANKCA2N5416	JANHCB2N5416 JANKCB2N5416

* 6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, MI
Air Force - 19, 99

Preparing activity:
DLA - CC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/485H	2. DOCUMENT DATE 10 January 2003

3. **DOCUMENT TITLE:** SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, LOW-POWER TYPES: 2N5415, 2N5415S, 2N5415UA, 2N5416, 2N5416S, AND 2N5416UA JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC.

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

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a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial 614-692-0510	DSN 850-0510	FAX 614-692-6939	EMAIL alan.barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus ATTN: DSCC-VAC P.O. Box 3990 Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533 Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888			

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